

# Physics 513 QFT HW 3

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### 1. 1

Let the spacetime translation operator be  $T(\vec{a})$  where  $a$  is the displacement 4-vector. I will show that the infinitesimal displacement operator is

$$T(\delta\vec{x}) = 1 - iP^\mu \delta x_\mu. \quad (1)$$

I show that it is unitary (like a translation operator must be).

$$T^\dagger(\delta\vec{x}) = 1 + iP^\mu \delta x_\mu. \quad (2)$$

$$\begin{aligned} T(\delta\vec{x})T^\dagger(\delta\vec{x}) &= (1 - iP^\mu \delta x_\mu)(1 + iP^\nu \delta x_\nu) = \\ &= 1 + \cancel{iP^\mu \delta x_\mu} - \cancel{iP^\mu \delta x_\mu} + P^\mu P^\nu \delta x_\mu \delta x_\nu = \\ &= 1 + \cancel{P^\mu P^\nu \delta x_\mu \delta x_\nu} = 1, \end{aligned} \quad (3)$$

where the higher order  $\delta x^2$  term goes to zero quickly enough to be ignored.

Now I show that it moves the field like it should

$$\phi(\vec{x} + \delta\vec{x}) = T^\dagger(\delta\vec{x})\phi(\vec{x})T(\delta\vec{x}). \quad (4)$$

I expand the left hand side with a Taylor series to first order

$$\phi(\vec{x} + \delta\vec{x}) = \phi(\vec{x}) + \delta x^\mu \partial_\mu \phi(\vec{x}). \quad (5)$$

$$\phi(\vec{x}) + \delta x^\mu \partial_\mu \phi(\vec{x}) = T^\dagger(\delta\vec{x})\phi(\vec{x})T(\delta\vec{x}) \quad (6)$$

$$\phi(\vec{x}) + \delta x^\mu \partial_\mu \phi(\vec{x}) = (1 + iP^\mu \delta x_\mu)\phi(\vec{x})(1 - iP^\nu \delta x_\nu) \quad (7)$$

$$\phi(\vec{x}) + \delta x^\mu \partial_\mu \phi(\vec{x}) = (1 + iP^\mu \delta x_\mu)\phi(\vec{x})(1 - iP^\nu \delta x_\nu) \quad (8)$$

$$\phi(\vec{x}) + \delta x^\mu \partial_\mu \phi(\vec{x}) = (1 + iP^\mu \delta x_\mu)[\phi(\vec{x}) - i\phi(\vec{x})P^\nu \delta x_\nu] \quad (9)$$

$$\begin{aligned} \phi(\vec{x}) + \delta x^\mu \partial_\mu \phi(\vec{x}) &= \phi(\vec{x}) + i[P^\mu \phi(\vec{x})\delta x_\mu - \phi(\vec{x})P^\nu \delta x_\nu] \\ &\quad + \cancel{P^\mu \phi(\vec{x})P^\nu \delta x_\mu \delta x_\nu} \end{aligned} \quad (10)$$

$$\phi(\vec{x}) + \delta x^\mu \partial_\mu \phi(\vec{x}) = \phi(\vec{x}) + i[P^\mu \phi(\vec{x}) - \phi(\vec{x})P^\mu]\delta x_\mu \quad (11)$$

$$\delta x^\mu \partial_\mu \phi(\vec{x}) = i[P^\mu \phi(\vec{x}) - \phi(\vec{x})P^\mu]\delta x_\mu \quad (12)$$

$$\delta x^\mu \partial_\mu \phi(\vec{x}) = i[P^\mu, \phi(\vec{x})]\delta x_\mu \quad (13)$$

$$\delta x^\mu \partial_\mu \phi(\vec{x}) = i[P^\mu, \phi(\vec{x})]\delta x_\mu \quad (14)$$

$$\partial^\mu \phi(\vec{x}) = i[P^\mu, \phi(\vec{x})] \quad (15)$$

But this is just the Heisenberg equation of motion and space translation, i.e. Eq. (1) in the problem set. If it is true then  $T(\delta\vec{x})$  is the infinitesimal spacetime translation operator.

To get the finite translation operator I compound infinitesimal translations

$$T(\vec{a}) = \lim_{N \rightarrow \infty} T(\delta\vec{a})^N, \quad \delta\vec{a} = \frac{\vec{a}}{N}. \quad (16)$$

$$T(\vec{a}) = \lim_{N \rightarrow \infty} \left(1 - iP^\mu \frac{a_\mu}{N}\right)^N = \exp(-iP^\mu a_\mu). \quad (17)$$

Thus the spacetime translation operator is

$$T(\vec{a}) = \exp(-iP^\mu a_\mu), \quad (18)$$

which acts on scalar fields by

$$T^\dagger(\vec{a})\phi(\vec{x})T(\vec{a}) = \phi(\vec{x} + \vec{a}). \quad (19)$$

QED.

### 2. 2

$$\phi(\mathbf{r}) = \sum_{\mathbf{p}} \frac{a(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{r}} + b^\dagger(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{2Vp^t}} \quad (20)$$

Field operators in the Schrödinger picture are functions of space (not space and time). The spacial position is

$$\mathbf{r} = (x, y, z). \quad (21)$$

The 4-momentum is

$$\vec{p} = (p^t, \mathbf{p}), \quad \mathbf{p} = (p^x, p^y, p^z). \quad (22)$$

I prefer to write  $\phi(\mathbf{r})$  as it appears in Peskin & Schroeder

$$E = p^t = \sqrt{p^2 + m^2} = \omega(p), \quad p^2 = \delta_{ij} p^i p^j. \quad (23)$$

$$\phi(\mathbf{r}) = \sum_{\mathbf{p}} \frac{a(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{r}} + b^\dagger(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{2V\omega(p)}} \quad (24)$$

$V$  is the volume of the factious box which keeps everything finite and normalizable. I take the thermodynamic limit

$$\lim_{V \rightarrow \infty} \sum_{\mathbf{p}} = \frac{V}{(2\pi)^3} \int d^3 p. \quad (25)$$

$$\phi(\mathbf{r}) = \int d^3 p \frac{a(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{r}} + b^\dagger(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}}}{(2\pi)^3 \sqrt{2\omega(p)}} \quad (26)$$

$$\mathbf{p} \cdot \mathbf{r} = \delta_{ij} p^i r^j = p^x x + p^y y + p^z z \quad (27)$$

$$\phi^\dagger(\mathbf{r}) = \int d^3 p \frac{a^\dagger(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}} + b(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{r}}}{(2\pi)^3 \sqrt{2\omega(p)}} \quad (28)$$

$$\phi(\mathbf{r}) = \frac{\phi_1(\mathbf{r}) + i\phi_2(\mathbf{r})}{\sqrt{2}}, \quad \phi^\dagger(\mathbf{r}) = \frac{\phi_1(\mathbf{r}) - i\phi_2(\mathbf{r})}{\sqrt{2}} \quad (29)$$

The scalar fields  $\phi_1(\mathbf{r})$  and  $\phi_2(\mathbf{r})$  are real. I know how to write real valued scalar fields in terms of creation & annihilation operators.

$$\phi_n(\mathbf{r}) = \int d^3 p \frac{a_n(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{r}} + a_n^\dagger(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}}}{(2\pi)^3 \sqrt{2\omega(p)}}, \quad n = 1, 2 \quad (30)$$

From my notes on real scalar fields I know the commutation relations these operators obey

$$\begin{aligned} [a_n(\mathbf{p}), a_m(\mathbf{k})] &= [a_n^\dagger(\mathbf{p}), a_m^\dagger(\mathbf{k})] = 0, \\ [a_n(\mathbf{p}), a_m^\dagger(\mathbf{k})] &= (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{k}) \delta_{nm}, \end{aligned} \quad (31)$$

where  $n, m = 1, 2$ .

$$\sqrt{2}\phi(\mathbf{r}) = \phi_1(\mathbf{r}) + i\phi_2(\mathbf{r}) \quad (32)$$

$$\begin{aligned} \int d^3 p \frac{a(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{r}} + b^\dagger(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{\omega(p)}} &= \\ \int d^3 p \frac{a_1(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{r}} + a_1^\dagger(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{2\omega(p)}} &+ \\ i \int d^3 p \frac{a_2(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{r}} + a_2^\dagger(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{2\omega(p)}} & \end{aligned} \quad (33)$$

I cannot prove it but it is very reasonable to assume that this equation holds for every mode, i.e. the integrands are equal.

$$\begin{aligned} a(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{r}} + b^\dagger(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}} &= \\ \frac{[a_1(\mathbf{p}) + ia_2(\mathbf{p})]e^{-i\mathbf{p}\cdot\mathbf{r}} + [a_1^\dagger(\mathbf{p}) + ia_2^\dagger(\mathbf{p})]e^{i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{2}} & \end{aligned} \quad (34)$$

I group by like terms.

$$a(\mathbf{p})e^{-i\mathbf{p}\cdot\mathbf{r}} = \frac{[a_1(\mathbf{p}) + ia_2(\mathbf{p})]e^{-i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{2}} \quad (35)$$

$$\boxed{\sqrt{2}a(\mathbf{p}) = a_1(\mathbf{p}) + ia_2(\mathbf{p})} \quad (36)$$

QED

$$\sqrt{2}b^\dagger(\mathbf{p}) = a_1^\dagger(\mathbf{p}) + ia_2^\dagger(\mathbf{p}) \quad (37)$$

$$\boxed{\sqrt{2}b(\mathbf{p}) = a_1(\mathbf{p}) - ia_2(\mathbf{p})} \quad (38)$$

QED

I now derive the nontrivial (trivial meaning ones like  $[a, a] = 0$ ) requested commutation relations

$$[a(\mathbf{p}), a^\dagger(\mathbf{k})] = \left[ \frac{a_1(\mathbf{p}) + ia_2(\mathbf{p})}{\sqrt{2}}, \frac{a_1^\dagger(\mathbf{k}) - ia_2^\dagger(\mathbf{k})}{\sqrt{2}} \right] \quad (39)$$

$$2[a(\mathbf{p}), a^\dagger(\mathbf{k})] = [a_1(\mathbf{p}) + ia_2(\mathbf{p}), a_1^\dagger(\mathbf{k}) - ia_2^\dagger(\mathbf{k})] \quad (40)$$

$$[A + B, C] = [A, C] + [B, C] \quad (41)$$

$$\begin{aligned} 2[a(\mathbf{p}), a^\dagger(\mathbf{k})] &= [a_1(\mathbf{p}) + ia_2(\mathbf{p}), a_1^\dagger(\mathbf{k})] \\ &\quad - i[a_1(\mathbf{p}) + ia_2(\mathbf{p}), a_2^\dagger(\mathbf{k})] \end{aligned} \quad (42)$$

$$2[a(\mathbf{p}), a^\dagger(\mathbf{k})] = [a_1(\mathbf{p}), a_1^\dagger(\mathbf{k})] - i^2[a_2(\mathbf{p}), a_2^\dagger(\mathbf{k})] \quad (43)$$

$$2[a(\mathbf{p}), a^\dagger(\mathbf{k})] = [a_1(\mathbf{p}), a_1^\dagger(\mathbf{k})] + [a_2(\mathbf{p}), a_2^\dagger(\mathbf{k})] \quad (44)$$

$$\boxed{[a(\mathbf{p}), a^\dagger(\mathbf{k})] = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{k})} \quad (45)$$

$$\boxed{[b(\mathbf{p}), b^\dagger(\mathbf{k})] = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{k})} \quad (46)$$

$$2[a(\mathbf{p}), b(\mathbf{k})] = [a_1(\mathbf{p}) + ia_2(\mathbf{p}), a_1(\mathbf{k}) - ia_2(\mathbf{k})] \quad (47)$$

$$2[a(\mathbf{p}), b(\mathbf{k})] = [a_1(\mathbf{p}), a_1(\mathbf{k})] - i[a_2(\mathbf{p}), ia_2(\mathbf{k})] \quad (48)$$

$$[a(\mathbf{p}), b(\mathbf{k})] = 0 \quad (49)$$

$$2[a^\dagger(\mathbf{p}), b^\dagger(\mathbf{k})] = [a_1^\dagger(\mathbf{p}) - ia_2^\dagger(\mathbf{p}), a_1^\dagger(\mathbf{k}) + ia_2^\dagger(\mathbf{k})] \quad (50)$$

$$[a^\dagger(\mathbf{p}), b^\dagger(\mathbf{k})] = 0 \quad (51)$$

$$2[a(\mathbf{p}), b^\dagger(\mathbf{k})] = [a_1(\mathbf{p}) + ia_2(\mathbf{p}), a_1^\dagger(\mathbf{k}) + ia_2^\dagger(\mathbf{k})] \quad (52)$$

$$[a(\mathbf{p}), b^\dagger(\mathbf{k})] = 0 \quad (53)$$

$$2[a^\dagger(\mathbf{p}), b(\mathbf{k})] = [a_1^\dagger(\mathbf{p}) - ia_2^\dagger(\mathbf{p}), a_1(\mathbf{k}) - ia_2(\mathbf{k})] \quad (54)$$

$$[a^\dagger(\mathbf{p}), b(\mathbf{k})] = 0 \quad (55)$$

### 3. 3

$$\phi(\mathbf{r}) = \int d^3p \frac{a_{\mathbf{p}} e^{-i\mathbf{p}\cdot\mathbf{r}} + b_{\mathbf{p}}^\dagger e^{i\mathbf{p}\cdot\mathbf{r}}}{(2\pi)^3 \sqrt{2\omega_p}}, \quad (56)$$

$$\phi^\dagger(\mathbf{r}) = \int d^3p \frac{a_{\mathbf{p}}^\dagger e^{i\mathbf{p}\cdot\mathbf{r}} + b_{\mathbf{p}} e^{-i\mathbf{p}\cdot\mathbf{r}}}{(2\pi)^3 \sqrt{2\omega_p}}$$

$$\sqrt{2}(2\pi)^3 \phi(\mathbf{r}) = \int d^3p \frac{a_{\mathbf{p}} e^{-i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{\omega_p}} + \int d^3p \frac{b_{\mathbf{p}}^\dagger e^{i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{\omega_p}} \quad (57)$$

$$I = \int d^3p \frac{a_{\mathbf{p}} e^{-i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{\omega_p}} \quad (58)$$

$$\sqrt{2}(2\pi)^3 \phi(\mathbf{r}) = I + \int d^3p \frac{b_{\mathbf{p}}^\dagger e^{i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{\omega_p}} \quad (59)$$

$$I = \int d^3k \frac{a_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{\omega_k}} \quad (60)$$

I do a u-substitution  $\mathbf{k} = -\mathbf{p}$ .

$$\begin{aligned} I &= \int_{-\infty}^{\infty} dk^x \int_{-\infty}^{\infty} dk^y \int_{-\infty}^{\infty} dk^z \frac{a_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{\omega_k}} = \\ &= - \int_{-\infty}^{\infty} dp^x \int_{-\infty}^{\infty} dp^y \int_{-\infty}^{\infty} dp^z \frac{a_{-\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{\omega_p}} = \\ &= \int_{-\infty}^{\infty} dp^x \int_{-\infty}^{\infty} dp^y \int_{-\infty}^{\infty} dp^z \frac{a_{-\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{\omega_p}} = \int d^3p \frac{a_{-\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{r}}}{\sqrt{\omega_p}} \end{aligned} \quad (61)$$

$$\phi(\mathbf{r}) = \int d^3p \frac{a_{-\mathbf{p}} + b_{\mathbf{p}}^\dagger}{(2\pi)^3 \sqrt{2\omega_p}} e^{i\mathbf{p}\cdot\mathbf{r}} \quad (62)$$

$$\phi^\dagger(\mathbf{r}) = \int d^3p \frac{a_{-\mathbf{p}}^\dagger + b_{\mathbf{p}}}{(2\pi)^3 \sqrt{2\omega_p}} e^{-i\mathbf{p}\cdot\mathbf{r}} \quad (63)$$

$$\phi(\mathbf{r}) e^{-i\mathbf{p}\cdot\mathbf{r}} = \int d^3k \frac{a_{-\mathbf{k}} + b_{\mathbf{k}}^\dagger}{(2\pi)^3 \sqrt{2\omega_k}} e^{i(\mathbf{k}-\mathbf{p})\cdot\mathbf{r}} \quad (64)$$

$$\int d^3x \phi(\mathbf{r}) e^{-i\mathbf{p}\cdot\mathbf{r}} = \int d^3k \frac{a_{-\mathbf{k}} + b_{\mathbf{k}}^\dagger}{(2\pi)^3 \sqrt{2\omega_k}} \int d^3x e^{i(\mathbf{k}-\mathbf{p})\cdot\mathbf{r}} \quad (65)$$

In my scalar field notes I showed that

$$\int d^3x e^{i\mathbf{u}\cdot\mathbf{r}} = (2\pi)^3 \delta(\mathbf{u}). \quad (66)$$

$$\int d^3x \phi(\mathbf{r}) e^{-i\mathbf{p}\cdot\mathbf{r}} = \int d^3k \frac{a_{-\mathbf{k}} + b_{\mathbf{k}}^\dagger}{\sqrt{2\omega_k}} \delta(\mathbf{k} - \mathbf{p}) \quad (67)$$

$$\sqrt{2\omega_p} \int d^3x \phi(\mathbf{r}) e^{-i\mathbf{p}\cdot\mathbf{r}} = a_{-\mathbf{p}} + b_{\mathbf{p}}^\dagger \quad (68)$$

I take  $\mathbf{p} \rightarrow -\mathbf{p}$ .

$$\sqrt{2\omega_p} \int d^3x \phi(\mathbf{r}) e^{i\mathbf{p}\cdot\mathbf{r}} = a_{\mathbf{p}} + b_{-\mathbf{p}}^\dagger \quad (69)$$

To go farther I need the conjugate momentum density. To save time I cheat a little and get it from Chat GPT.

$$\pi(\mathbf{r}) = i \int \frac{d^3p}{(2\pi)^3} \sqrt{\frac{\omega_p}{2}} (a_{\mathbf{p}}^\dagger e^{i\mathbf{p}\cdot\mathbf{r}} - b_{\mathbf{p}} e^{-i\mathbf{p}\cdot\mathbf{r}}) \quad (70)$$

$$\pi(\mathbf{r}) = i \int \frac{d^3p}{(2\pi)^3} \sqrt{\frac{\omega_p}{2}} (a_{\mathbf{p}}^\dagger - b_{-\mathbf{p}}) e^{i\mathbf{p}\cdot\mathbf{r}} \quad (71)$$

$$\pi(\mathbf{r}) = i \int \frac{d^3k}{(2\pi)^3} \sqrt{\frac{\omega_k}{2}} (a_{\mathbf{k}}^\dagger - b_{-\mathbf{k}}) e^{i\mathbf{k}\cdot\mathbf{r}} \quad (72)$$

$$\begin{aligned} \int d^3x \pi(\mathbf{r}) e^{-i\mathbf{p}\cdot\mathbf{r}} &= i \int \frac{d^3k}{(2\pi)^3} \sqrt{\frac{\omega_k}{2}} (a_{\mathbf{k}}^\dagger - b_{-\mathbf{k}}) \\ &\quad \int d^3x e^{i(\mathbf{k}-\mathbf{p})\cdot\mathbf{r}} \end{aligned} \quad (73)$$

$$\int d^3x \pi(\mathbf{r}) e^{-i\mathbf{p}\cdot\mathbf{r}} = i \int d^3k \sqrt{\frac{\omega_k}{2}} (a_{\mathbf{k}}^\dagger - b_{-\mathbf{k}}) \delta^{(3)}(\mathbf{k} - \mathbf{p}) \quad (74)$$

$$-i\sqrt{\frac{2}{\omega_p}} \int d^3x \pi(\mathbf{r}) e^{-i\mathbf{p}\cdot\mathbf{r}} = a_{\mathbf{p}}^\dagger - b_{-\mathbf{p}} \quad (75)$$

$$a_{\mathbf{p}} = \frac{1}{\sqrt{2\omega_p}} \int d^3x [\omega_p \phi(\mathbf{r}) + i\pi^\dagger(\mathbf{r})] e^{i\mathbf{p}\cdot\mathbf{r}} \quad (90)$$

$$i\sqrt{\frac{2}{\omega_p}} \int d^3x \pi^\dagger(\mathbf{r}) e^{i\mathbf{p}\cdot\mathbf{r}} = a_{\mathbf{p}} - b_{-\mathbf{p}}^\dagger \quad (76)$$

$$C a_{\mathbf{p}} C^\dagger = C \frac{1}{\sqrt{2\omega_p}} \int d^3x [\omega_p \phi(\mathbf{r}) + i\pi^\dagger(\mathbf{r})] e^{i\mathbf{p}\cdot\mathbf{r}} C^\dagger \quad (91)$$

$$\boxed{a_{\mathbf{p}} = \frac{1}{\sqrt{2\omega_p}} \int d^3x [\omega_p \phi(\mathbf{r}) + i\pi^\dagger(\mathbf{r})] e^{i\mathbf{p}\cdot\mathbf{r}}} \quad (77)$$

$$C a_{\mathbf{p}} C^\dagger = \frac{1}{\sqrt{2\omega_p}} \int d^3x [\omega_p C \phi(\mathbf{r}) C^\dagger + i C \pi^\dagger(\mathbf{r}) C^\dagger] e^{i\mathbf{p}\cdot\mathbf{r}} \quad (92)$$

$$\sqrt{2\omega_p} \int d^3x \phi^\dagger(\mathbf{r}) e^{-i\mathbf{p}\cdot\mathbf{r}} = a_{\mathbf{p}}^\dagger + b_{-\mathbf{p}} \quad (78)$$

$$C a_{\mathbf{p}} C^\dagger = \frac{1}{\sqrt{2\omega_p}} \int d^3x [\omega_p \eta_c \phi^\dagger(\mathbf{r}) + i \eta_c \pi(\mathbf{r})] e^{i\mathbf{p}\cdot\mathbf{r}} \quad (93)$$

$$b_{-\mathbf{p}} = \frac{1}{\sqrt{2\omega_p}} \int d^3x [\omega_p \phi^\dagger(\mathbf{r}) + i\pi(\mathbf{r})] e^{-i\mathbf{p}\cdot\mathbf{r}} \quad (79)$$

$$C a_{\mathbf{p}} C^\dagger = \frac{\eta_c}{\sqrt{2\omega_p}} \int d^3x [\omega_p \phi^\dagger(\mathbf{r}) + i\pi(\mathbf{r})] e^{i\mathbf{p}\cdot\mathbf{r}} \quad (94)$$

$$\boxed{b_{\mathbf{p}} = \frac{1}{\sqrt{2\omega_p}} \int d^3x [\omega_p \phi^\dagger(\mathbf{r}) + i\pi(\mathbf{r})] e^{i\mathbf{p}\cdot\mathbf{r}}} \quad (80)$$

$$b_{\mathbf{p}} = \frac{1}{\sqrt{2\omega_p}} \int d^3x [\omega_p \phi^\dagger(\mathbf{r}) + i\pi(\mathbf{r})] e^{i\mathbf{p}\cdot\mathbf{r}} \quad (95)$$

The problem states  $C$  is unitary so  $C^\dagger = C^{-1}$ .

$$C C^\dagger = C^\dagger C = 1 \quad (81)$$

$$\boxed{C a_{\mathbf{p}} C^\dagger = \eta_c b_{\mathbf{p}}} \quad (96)$$

The problem states

$$\boxed{C \phi(\mathbf{r}) C^\dagger = \eta_c \phi^\dagger(\mathbf{r})} \quad (82)$$

QED

$$C b_{\mathbf{p}} C^\dagger = \frac{1}{\sqrt{2\omega_p}} \int d^3x [\omega_p C \phi^\dagger(\mathbf{r}) C^\dagger + i C \pi(\mathbf{r}) C^\dagger] e^{i\mathbf{p}\cdot\mathbf{r}} \quad (97)$$

$$[C \phi(\mathbf{r}) C^\dagger]^\dagger = C [C \phi(\mathbf{r})]^\dagger = C \phi^\dagger(\mathbf{r}) C^\dagger = \eta_c^* \phi(\mathbf{r}) \quad (83)$$

$$C b_{\mathbf{p}} C^\dagger = \frac{\eta_c^*}{\sqrt{2\omega_p}} \int d^3x [\omega_p \phi(\mathbf{r}) + i\pi^\dagger(\mathbf{r})] e^{i\mathbf{p}\cdot\mathbf{r}} \quad (98)$$

$$\boxed{C \phi^\dagger(\mathbf{r}) C^\dagger = \eta_c^* \phi(\mathbf{r})} \quad (84)$$

I am working in the Schrödinger picture. In the Heisenberg picture I assume that Eqs. 82 & 84 still hold

$$C \phi(\vec{x}) C^\dagger = \eta_c \phi^\dagger(\vec{x}), \quad C \phi^\dagger(\vec{x}) C^\dagger = \eta_c^* \phi(\vec{x}), \quad \vec{x} = (t, \mathbf{r}). \quad (85)$$

$$a_{\mathbf{p}} = \frac{1}{\sqrt{2\omega_p}} \int d^3x [\omega_p \phi(\mathbf{r}) + i\pi^\dagger(\mathbf{r})] e^{i\mathbf{p}\cdot\mathbf{r}} \quad (99)$$

I cheat again and get these relations from Chat GPT

$$\boxed{C b_{\mathbf{p}} C^\dagger = \eta_c^* a_{\mathbf{p}}} \quad (100)$$

$$\pi(\vec{x}) = \dot{\phi}^\dagger(\vec{x}), \quad \pi^\dagger(\vec{x}) = \dot{\phi}(\vec{x}). \quad (86)$$

QED

Then

$$\begin{aligned} \pi(\vec{x}) &= \dot{\phi}^\dagger(\vec{x}) = \partial_t \phi^\dagger(\vec{x}) \implies C \pi(\vec{x}) C^\dagger = C [\partial_t \phi^\dagger(\vec{x})] C^\dagger \\ &= \partial_t C \phi^\dagger(\vec{x}) C^\dagger = \partial_t \eta_c^* \phi(\vec{x}) = \eta_c^* \partial_t \phi(\vec{x}) = \eta_c^* \dot{\phi}(\vec{x}) = \eta_c^* \pi^\dagger(\vec{x}), \end{aligned} \quad (87)$$

$$C \pi(\vec{x}) C^\dagger = \eta_c^* \pi^\dagger(\vec{x}), \quad C \pi^\dagger(\vec{x}) C^\dagger = \eta_c \pi(\vec{x}). \quad (88)$$

I assume these relations also hold in the Schrödinger picture

$$\boxed{C \pi(\mathbf{r}) C^\dagger = \eta_c^* \pi^\dagger(\mathbf{r}), \quad C \pi^\dagger(\mathbf{r}) C^\dagger = \eta_c \pi(\mathbf{r})} \quad (89)$$

$$|\mathbf{p}_a\rangle = a_{\mathbf{p}}^\dagger |0\rangle, \quad |\mathbf{p}_b\rangle = b_{\mathbf{p}}^\dagger |0\rangle \quad (101)$$

$$C |\mathbf{p}_a\rangle = C a_{\mathbf{p}}^\dagger |0\rangle = C a_{\mathbf{p}}^\dagger C^\dagger C |0\rangle \quad (102)$$

$$C |0\rangle = |0\rangle \quad (103)$$

$$C |\mathbf{p}_a\rangle = C a_{\mathbf{p}}^\dagger |0\rangle = C a_{\mathbf{p}}^\dagger C^\dagger |0\rangle \quad (104)$$

$$C a_{\mathbf{p}} C^\dagger = \eta_c b_{\mathbf{p}} \implies C a_{\mathbf{p}}^\dagger C^\dagger = \eta_c^* b_{\mathbf{p}}^\dagger \quad (105)$$

$$C |\mathbf{p}_a\rangle = C a_{\mathbf{p}}^\dagger C^\dagger |0\rangle = \eta_c^* b_{\mathbf{p}}^\dagger |0\rangle = \eta_c^* |\mathbf{p}_b\rangle \quad (106)$$

$$\boxed{C |\mathbf{p}_a\rangle = \eta_c^* |\mathbf{p}_b\rangle} \quad (107)$$

$$C |\mathbf{p}_b\rangle = C b_{\mathbf{p}}^\dagger C^\dagger C |0\rangle = C b_{\mathbf{p}}^\dagger C^\dagger |0\rangle \quad (108)$$

$$C b_{\mathbf{p}} C^\dagger = \eta_c^* a_{\mathbf{p}} \implies C b_{\mathbf{p}}^\dagger C^\dagger = \eta_c a_{\mathbf{p}}^\dagger \quad (109)$$

$$C |\mathbf{p}_b\rangle = C b_{\mathbf{p}}^\dagger C^\dagger |0\rangle = \eta_c a_{\mathbf{p}}^\dagger |0\rangle = \eta_c |\mathbf{p}_a\rangle \quad (110)$$

$$\boxed{C |\mathbf{p}_b\rangle = \eta_c |\mathbf{p}_a\rangle} \quad (111)$$

QED